



RUSSIAN TORTOISE · TESTUDO HORSFIELDII

Keep your tortoise the right way for forty years.

A complete, fact-checked printable manual: enclosure size and why a tank is the wrong shape, ultraviolet B (UVB) light you can actually measure, a steppe grazer's real diet, the brumation decision, and the health pages for an animal that will outlast most of your furniture.

INSIDE

- ✓ Enclosure sizing: the minimum floor, the target, and why a tank is the wrong shape
- ✓ A weed list, a grazing plan, and the never-feed list for an arid grazer
- ✓ The brumation decision, in full, with the protocol if you say yes
- ✓ Seven health pages, the law in every state, and fillable owner and sitter sheets

51 pages

Intermediate

Print or tablet

Edition 3.1 · Oct 2026



CONTENTS

What's inside

Every page is written to be used, not just read once.

GETTING STARTED

How to Use This Package	3
-------------------------	---

SECTION 01 · PROFILE

Quick Profile & Cost Overview	4
The Species, Size & Lifespan	5

SECTION 02 · HOUSING & ENVIRONMENT

Enclosure Size, Type & Where It Goes	6
Temperature, Basking & Night Lows	7
Thermostats, Timers & Thermometers	8
UVB: Tube, Strength & Mounting Distance	9
Humidity, Substrate & Furnishings	10
Outdoor Pens & Escape-Proofing	11
Cleaning & Hygiene	12

SECTION 03 · FEEDING

Diet by Age & How Much	13
Weeds, Hay & Growing Your Own	14
Fruit, Protein & the Never-Feed List	15
Calcium, Multivitamin & Water	16
Growth, Weight & Body Condition	17

SECTION 04 · HANDLING & BEHAVIOR

Handling & Hygiene	18
Daily Rhythm, Courtship & Living Alone	19
Enrichment & Common Mistakes	20

SECTION 05 · ARRIVAL & LIFE STAGES

Choosing a Tortoise, Quarantine & the First Vet Visit	21
Sexing & Females That Lay	22
Egg Binding	23
Brumation: the Decision & the Protocol	24

SECTION 06 · THE LAW

Is a Russian Tortoise Legal Where You Live?	25
The Four-Inch Rule & the Trade	26

SECTION 07 · HEALTH & COMMON ISSUES

Red Flags & Finding a Vet	27
Metabolic Bone Disease & Pyramiding	28
Respiratory Infection & Herpesvirus	29
Shell Rot, Shell Trauma & Abscesses	30
Parasites & Hexamita	31
Vitamin A, Bladder Stones & Beak Overgrowth	32
Reading Droppings, Urates & Hydration	33

SECTION 08 · QUICK REFERENCE

Setup Checklist & Targets	34
Emergency & Quick Targets Card	35

SECTION 09 · OWNER TOOLS

Budget: Setup & Shopping List	36
Budget: Yearly Costs	37
First 30 Days	38
Symptom Quick Reference	39
Daily, Weekly & Seasonal Routine	40
Power Outages, Heat Waves, Travel & Transport	41
Pet-Sitter Sheet	42
Owner Log	43
Equipment & Vet Log	44
Enrichment Checklist & Log	45

REFERENCE

Glossary, A to M	46
Glossary, N to Z	47
Sources	48
Sources, Continued	49
Where the Sources Disagree	50
Version History & About	51

How to use this package

Three ways to get value from it, depending on where you are right now.

UNITS USED THROUGHOUT

Every temperature and size is given twice. Temperatures appear in degrees Fahrenheit (°F) with degrees Celsius (°C) beside them. Lengths appear in inches (in) and feet (ft), with centimeters (cm), millimeters (mm) or meters (m) beside them, and floor area in square feet (sq ft). A tortoise and its weight log are measured in grams (g), the unit a kitchen scale shows, with pounds (lb) for scale. Soaks are timed in minutes (min). Humidity is a percentage (%) of the moisture the air could hold. Prices are in United States dollars.

If you haven't bought a tortoise yet

Check the law first, on page 25: a Russian tortoise is legal in most of the United States, but one state bans it and five want a permit. Then read pages 4 to 12 in order before buying anything. The enclosure and the UVB fixture are the two decisions people get wrong most, and both are expensive to redo. Buy once, from the Budget & Shopping List on page 36.

If your tortoise is already home

Audit what you have against the Setup Checklist on page 34. Do two checks today: the basking spot's surface temperature, read with the infrared temperature gun, and the UVB tube's age, since its output fades long before the light looks dim. If the animal itself seems off, go to page 27.

Print these first

- | | |
|--|---|
| ☐ Page 34: Setup Checklist & Targets | ☐ Page 42: Pet-Sitter Sheet |
| ☐ Page 35: Emergency & Quick Targets Card | ☐ Page 43: Owner Log |
| ☐ Page 39: Symptom Quick Reference | ☐ Page 44: Equipment & Vet Log |
| ☐ Page 40: Daily, Weekly & Seasonal Routine | ☐ Page 15: the never-feed list, for the fridge |

WHY THIS PACKAGE EXISTS

The Russian tortoise is hardy, which is why poor care can run for years before it shows. Every figure here is cross-checked against the veterinary and specialist sources on page 48.

NOT A SUBSTITUTE FOR

A reptile veterinarian with tortoise experience. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a vet call, not a page in this book. Find a reptile vet before you need one and write the number on the emergency card. Nothing on the law pages is legal advice either.

SECTION 01 • PROFILE

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Testudo horsfieldii</i> , also sold as Horsfield's tortoise
Native range	Central Asia: Kazakhstan, Uzbekistan, Iran, Afghanistan, Pakistan and western China. Semi-arid steppe, rocky desert and dry scrub
Adult size	5 to 10 in (13 to 25 cm) of shell; females run slightly larger than males
Captive lifespan	40 years or more is typical, and some reach 50 plus
Diet type	Herbivore: grass hay and weeds, high in fiber and calcium, no fruit and no animal protein (page 13)
Indoor enclosure	An open-topped tortoise table. 4×2 ft (120×60 cm), about 8 sq ft, is the floor; 7×3.5 ft (210×105 cm), about 24.5 sq ft, is the adult working size
Temperatures	Basking surface about 95°F (35°C); cool end in the 70s°F (21 to 26°C); night no lower than 60°F (16°C)
UVB	A linear T5 HO tube (T5: a fluorescent tube five-eighths of an inch across; HO: high output), giving a UVI (ultraviolet index, the strength of the UVB) of 3.0 to 4.0 at the basking spot
Humidity	40 to 50% at the open surface for an adult, 40 to 75% for a juvenile, plus a moist hide
Brumation	A winter slowdown the species is built for, and optional indoors (page 24)
Social	Alone, or two females with care. Never two males
Conservation status	Vulnerable on the IUCN (International Union for Conservation of Nature) Red List, the global list of threatened species, and on Appendix II of CITES (the Convention on International Trade in Endangered Species), so international trade runs on export quotas

Cost overview

Initial setup

\$50 to \$150 for a hatchling or young captive-bred tortoise, \$225 to \$400 for an older one. \$410 to \$835 for the full indoor setup, of which the table alone is \$125 to \$250. Every line is on page 36.

Yearly ongoing

\$365 to \$715 a year, about \$30 to \$60 a month, for food, supplement top-ups, substrate, the yearly UVB tube, electricity and a wellness exam with a fecal test (a stool check for parasites).

THE NUMBER THAT SHOULD DECIDE THIS

Forty years at \$365 to \$715 a year comes to \$14,600 to \$28,600, before a single illness, and illness with tests or medication commonly runs \$150 to \$300 or more each time. A tortoise bought today is very likely to need a second keeper. Plan who that is before you buy (page 41).

The species, size & lifespan

What you are buying, how big it gets, and how long it stays.

One species, several names

The pet Russian tortoise is *Testudo horsfieldii*, also sold as Horsfield's tortoise. It is a compact, hardy animal of the arid steppe and rocky desert of Central Asia, and one of the most cold-tolerant tortoises in the world. In the clay and sand deserts of its range it is above ground for only about three months a year, roughly March through May. It digs tunnels up to 2 m (about 6.5 ft) long into hillsides or under stones, or takes over an old marmot or hedgehog burrow, and from late May it drops out of sight: some populations spend June to August underground in a summer dormancy, and all of them sit out the winter deep in the same burrow.

Even in the active season it feeds in two short windows a day, mid-morning and late afternoon, and over a season wild tortoises eat more than 80 plant species from 27 families. That is the animal this package is built around: a burrow to disappear into, a daily round between shelter and forage, and a diet that changes constantly.

Size and growth

Adult length	5 to 10 in (13 to 25 cm) of shell. The largest verified animals reach about 22 cm (close to 9 in) of straight shell length.
Hatchling	About 42 to 50 mm (1.7 to 2 in) of shell and 18 to 23 g out of the egg
Growth	Slow, over years. Fast growth is a warning rather than an achievement (page 17)
Males and females	Males have a much longer tail, usually tucked to one side; females are slightly larger. Sexual maturity comes at around 10 years

Lifespan

40 years or more is typical in captivity, and some individuals reach 50 plus. Wild lifespan is not well documented; the widely quoted 40 to 50 years is a captive-care figure. A Russian tortoise stays small enough at 5 to 10 in to spend that whole life indoors on a tortoise table, which is its real advantage over the larger tortoises. It is also why rehoming is common: the animal routinely outlives the plans made for it.

WILD-CAUGHT, CAPTIVE-BRED, AND THE DIFFERENCE TO YOU

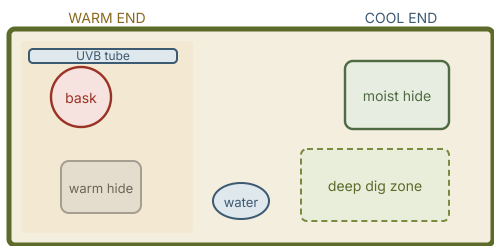
The species has been traded heavily for the pet market since the mid-1970s, and for most of that history a wild-caught animal was the norm. Uzbekistan is the main exporter today, Ukraine supplies a smaller share, and captive breeding runs alongside a continuing wild-caught supply. Wild-caught tortoises arrive stressed and carrying parasites, and die at a higher rate than captive-bred ones. Captive-bred costs more and is the better choice. What to look for in the animal itself is on page 21.

Enclosure size, type & where it goes

Floor area, an open top, and solid walls. Height hardly matters.

Floor area is the number that matters, because this is an unusually active tortoise that walks its whole enclosure every day and rearranges anything light enough to move. A 4×2 ft (120×60 cm) table, about 8 sq ft, is the absolute indoor minimum for one adult, and it is a floor rather than a goal: below it the enclosure is failing, and at it the job is not finished. Aim for at least 12 sq ft, and treat 7×3.5 ft (210×105 cm), about 24.5 sq ft, as the adult working size. Each extra tortoise needs at least 10 sq ft more.

WHO	FLOOR
Juvenile under about 4 in (10 cm) of shell	4×2 ft (120×60 cm) is enough until it reaches about 6 in (15 cm)
One adult, the floor	4×2 ft (120×60 cm), about 8 sq ft
One adult, the target	12 sq ft at least, and ideally 7×3.5 ft (210×105 cm), about 24.5 sq ft
Each added tortoise	At least 10 sq ft more, with cover that breaks every sightline



Seen from above. The basking lamp and the UVB tube sit together over the warm end, so the tortoise gets heat and UVB in the same place. The cool end stays genuinely cool, with the moist hide and a deeper dig zone. Hides, stones and dishes sit away from the walls.

An open table, not a glass tank

An open-topped tortoise table is strongly preferred over a closed glass aquarium, for two separate reasons. Glass ventilates poorly, and damp air held against an arid-adapted animal is a cause of respiratory infection. And a tortoise that can see through a wall it cannot pass tends to pace along it. Neither problem is fixed by buying a bigger tank.

- **Walls solid and smooth**, at least a foot (30 cm) high. Wire or mesh invites climbing, and this species climbs better than its shape suggests.
- **Nothing makes a ladder**. Hides, rocks, dishes and heaped substrate stay away from the walls, and corners are checked for a foothold.
- **Open on top indoors**, with room above for the lamps at their mounting distances and no shelf or lid in the light path.

Where it goes

Pick a room that holds at least 60°F (16°C) at night, or plan for a ceramic heater on the enclosure. Keep it out of drafts, which are one of the triggers for respiratory infection, and out of direct window sun: window glass blocks the UVB, so the sun through it adds heat and nothing the shell needs.

Temperature, basking & night lows

A real gradient, measured on the surface, in an unusually cold-hardy species.

A tortoise cannot make its own heat, so the enclosure offers a range and lets the animal choose: a hot basking spot at one end and a genuinely cool end at the other. If the whole table reads the same, the animal cannot regulate itself, whether that reading is too high or right on target.

ZONE	TARGET	HOW TO READ IT
Basking surface	About 95°F (35°C)	Infrared gun, on the surface
Air across the enclosure, by day	About 75 to 85°F (24 to 29°C)	The cool-end probe, moved to mid-table for a spot check
Cool end	The 70s°F (21 to 26°C)	Digital probe thermometer at the cool end
Night	A drop to about 60°F (16°C) is fine	The coldest reading of the night
Act below	60°F (16°C)	Indoors and out, outside brumation
Brumation, if chosen	35 to 50°F (2 to 10°C), ideally 41°F (5°C)	Page 24

A cold-hardy species, within limits

Russian tortoises are one of the most cold-tolerant tortoises kept, and a night drop to around 60°F (16°C) needs no added heat in most homes. Sixty is still the floor. A room that will not hold it needs a ceramic heater on a thermostat, since a heat source that gives off light is wrong at night.

The heat source

- **A halogen basking bulb** over the warm end is the daytime heat source indoors. Tortoises kept outdoors in mild climates often need no added heat at all.
- **Distance is the adjustment.** Raise or lower the lamp to set the surface temperature, then hold it there with a dimming thermostat. Wattage is a starting guess you verify, not a number to trust.
- **Keep every lamp at least 4 to 6 in (10 to 15 cm) above the highest point the tortoise can reach,** including the top of a hide or a rock it can climb.
- **Basking lamp and UVB tube together,** at the same end, so heat and ultraviolet light arrive in the same place, as they do in sunlight.

TWO THERMOMETERS, TWO JOBS

The infrared gun reads the basking surface; the digital probe thermometer reads the cool end. They are measuring different places on purpose and should read differently. A gradient is the goal, not a matching pair of numbers.

IN SUMMER, HEAT IS THE DANGER

Overheating is one of the five ways an outdoor tortoise dies (page 11). Indoors, check the enclosure on the hottest afternoon of the year, not the morning, and keep it out of a sunroom and out of direct sun through glass.

Thermostats, timers & thermometers

The equipment that holds the targets, and the three items first setups leave out.

A heat source with no thermostat draws full power with no ceiling, and a basking surface under it can climb well past what a reptile can tolerate. A thermostat measures the temperature at one point and cuts or lowers power once that point reaches the set value. Three items on the shopping list, the dimming thermostat, the infrared temperature gun and the gram scale, are the ones most first setups skip, and they are exactly the equipment that prevents thermal burns and catches slow weight loss.

CONTROLLER	WHAT IT DOES	SUITS
Dimming thermostat	Lowers the power to a bulb instead of switching it off and on	The halogen basking bulb. The right choice here
Pulse proportional	Sends short pulses of power to hold a steady average	A ceramic heater for night heat
On/off	Cuts power fully at the set point, restores it fully below it	Slow heat sources only, never a basking bulb

Where the probe goes

The probe sits where the tortoise is, at the basking spot, never up near the fixture reading room air. Makers disagree on whether it belongs on the basking surface itself or in the air a few inches below the bulb, so follow your own controller's instructions and then verify the result with the infrared gun. Hold the probe in place with a probe clip or a suction cup, never tape. A probe that slips and reads the wrong spot is how a correctly set thermostat still produces a burn.

The rest of the kit

- **A timer** on the lights, so the day length holds whether or not you are home. The cheapest item on the list.
- **A 6500 K daylight lamp** alongside the UVB tube, for bright white light over the warm end.
- **An infrared temperature gun** for the basking surface. It reads a surface from a distance, which no other instrument here does.
- **A digital probe thermometer** at the cool end, plus a **hygrometer** for humidity.
- **A digital scale reading in grams**, for the weight log that runs for the tortoise's whole life.
- **A UVI meter**, optional, the only way to measure the UVB at the basking spot rather than assume it.

HOW THERMOSTATS FAIL

Most fail obviously: the heat stops and the enclosure cools. The dangerous failure is rarer, a thermostat stuck on, or a probe that has slipped off its spot. A second, independent reading checked every day, the infrared gun on the basking surface, catches either one before it becomes a burn or a cold snap.

UVB: tube, strength & mounting distance

The one piece of kit you cannot substitute, and the one that fails invisibly.

UVB on the skin is how a tortoise makes vitamin D3, and vitamin D3 is how it absorbs calcium to build and keep a shell. A young tortoise short on UVB grows a soft, deformed shell, and that damage is permanent. The difficulty is that UVB output falls away well before the visible light dims, so a tube that is obviously working can be delivering very little.

REQUIREMENT	DETAIL
Tube	A linear T5 HO tube carrying the maker's 10 or 12% UVB rating printed on it, in a reflective fixture. Coil and compact UVB bulbs are not adequate for this species
Length and position	A third to half the enclosure length, over the warm end alongside the basking lamp
Target	A UVI of 3.0 to 4.0 at the basking area
Mounting distance	With mesh between tube and tortoise, its back sits 14 to 16 in (35 to 40 cm) below the tube. With no mesh, 17 to 20 in (43 to 50 cm)
Replacement	Every 6 to 12 months depending on the model, never later than 12, on the calendar rather than when it looks dim. Write the date on the equipment log (page 44)
Visible light	A bright 6500 K daylight lamp alongside. K, kelvin, is the scale for the color of light; 6500 K is white daylight
Day length	On a timer: 12 hours on and 12 off, or a seasonal swing from about 14 hours in summer to 10 in winter

Three ways to deliver no UVB while believing you are

- **The wrong bulb.** A coil or compact bulb is not a substitute for a linear T5 HO tube over a tortoise.
- **Glass or plastic in the path.** Window glass blocks essentially all UVB, and most clear plastics block all of it or close to it. A sunny window gives heat and light, not UVB. A mesh screen costs some output, which is what the two mounting distances above allow for.
- **A tube past its date.** Output decays from the day it is switched on. Distance matters too: UVB falls off sharply with distance, which is why the distances above are measured from the tortoise's back, not the floor, and why a few inches too high costs a large share of the output.

SUNLIGHT BEATS EVERY BULB

Unfiltered natural sunlight outdoors, in weather that suits the species, delivers UVB no lamp matches, and as little as 1 to 2 hours a week of it can have a marked effect on a reptile's health. It does not replace the indoor tube, because the weather does not cooperate for most of the year. Doing it safely is the subject of page 11.

Humidity, substrate & furnishings

An arid species that still needs somewhere damp, and soil deep enough to dig in.

Two humidities, not one

Keep the air at the open surface relatively low, 40 to 50% for an adult. A juvenile runs moister, 40 to 75%, lower by day and higher at night. High humidity across the whole enclosure is a real cause of respiratory infection, so resist the instinct to keep everything damp. But the tortoise lives in two climates: a dry, well-ventilated surface it walks and basks on, and a genuinely humid microclimate underneath, in a burrow or a moist hide, where it spends much of its day and does most of its shedding. Build both. A hide lined with damp peat moss on the cool side is the minimum. An enclosure that is dry everywhere, with no damp retreat and nothing to dig into, produces a dehydrated tortoise, and that is a more common failure in this species than one that is too wet.

Substrate

USE	NOTES
50/50 topsoil and coconut coir	The standard mix, kept slightly damp at depth rather than bone dry
Or a sand and loam blend	A similar soil mix that holds a burrow
Depth	At least 4 in (10 cm) for a juvenile and 6 in (15 cm) for an adult, deeper again in one dig zone
Dig zone	6 to 12 in (15 to 30 cm) or more in at least one part of the enclosure, or a dedicated dig box if the main bed cannot go that deep
Avoid: pure sand	Linked to mobility problems and impaction
Avoid: rabbit pellets, newspaper alone	Nothing to dig, and no support for natural behavior

DIGGING IS CLOSE TO A WELFARE NEED

Burrows are how this species buffers heat, cold and dryness in the wild, so an enclosure with no diggable zone is missing something the tortoise actively needs, not a nice extra.

Furnishings

- **At least two hides**, facing different directions: one warm and dry, one cool and moist. Terracotta half-pots, wooden hide boxes filled with loose substrate, cork flats, or a partly buried hide the tortoise can dig out further all work.
- **Terrain**: hollow logs, flat stones, low platforms and gentle slopes, which get used rather than walked past, and which break the space into routes. All of it away from the walls.
- **A water dish**: a shallow bowl, something like a plant saucer, sunk so its lip sits level with the surface. It doubles as a soaking dish.
- **A feeding tile or flat stone**, so food is not eaten off the soil.

Outdoor pens & escape-proofing

The richest setup this species can have, and the one it will try hardest to leave.

Whenever local temperature and security allow, a secure outdoor pen is the best enrichment there is for this species: real sunlight, real soil to dig, varied ground and plants growing where it walks. A pen shaped around the species is hilly and rocky, drains well, has shrubby cover such as lavender, rosemary and cistus, and is sown with an edible wildflower mix so the tortoise grazes what grows there. But a pen that is not dig-proof, climb-proof and predator-proof is a bigger risk than staying indoors. Wild Russian tortoises dig burrows up to 2 m (about 6.5 ft) long.

ELEMENT	WHAT IT NEEDS
Walls	Solid and opaque, never mesh or chain link, about 2 ft (60 cm) high with a lip of about 4 in (10 cm) projecting inward along the top
Buried barrier	Walls sunk at least 12 in (30 cm) into the ground, in concrete or buried wire mesh
Top	Covered with mesh. Dogs, foxes and raccoons come in at ground level, crows, ravens and birds of prey from above
Shade	Permanent: a solid roof section or a shelter, not a shadow that moves off by mid-afternoon
Shelter	An enclosed house that stays dry, with runoff draining away from the entrance, checked after storms
Water	Never deeper than the tortoise's head is tall, with a slope or ramp out from every side it can enter
Plants	Every plant in or reachable from the pen checked before the tortoise goes in (page 15)

The five ways an outdoor tortoise dies

- **Overheating.** A desert tortoise, a species built for heat, starts drooling heavily at a body temperature of 103°F (39.5°C), and 109°F (43°C) is lethal: under 7°F between the two. A tortoise drooling or frothing goes into shade at once and to a vet the same day.
- **Predation.** In Segura and colleagues' 2020 study of wild spur-thighed tortoises, a related species, young tortoises under 75 mm (about 3 in) of shell were taken by ravens, which caused every recorded death on open ground with little cover. Anything that small needs a covered pen and cover to duck under.
- **Escape.** Digging under or climbing a corner. Walk the perimeter regularly and look for fresh digging along the base.
- **Drowning.** A pool or dish it can get into and not out of, or an upended tortoise in water.
- **An unsafe plant** growing in or over the pen.

WHEN IT COMES BACK IN

The floor outdoors is the same 60°F (16°C) as indoors, but a 60°F afternoon usually means a night in the low fifties with wind and damp on top. Bring the tortoise in while the daytime reading is still comfortably above the floor, and act on the forecast rather than the afternoon thermometer.

Cleaning & hygiene

A little every day, a scrub when something is soiled, and a full change of soil a few times a year.

Cleaning and disinfecting are two different jobs. Cleaning is the physical removal of droppings, food and grime; disinfecting kills what is left on the surface afterward, and it only works on a surface that has already been scrubbed, because a film of dirt shields bacteria from the disinfectant.

WHEN	WHAT TO DO
Daily	Pick out droppings and urates (the white part of the waste) as soon as you see them, and any uneaten fresh food. Empty, wash and refill the water dish.
As it happens	Lift out any soiled patch of substrate and replace it.
Weekly	Wipe down dishes, hides and the feeding tile. Dampen the moist hide if it has dried out.
Every 3 to 6 months	Move the tortoise to a secure temporary tub, take everything out, scrub and disinfect the table, hides and dishes, and refill with all-new substrate.

Disinfecting safely

- Household bleach at 1:32, which is half a cup of bleach per gallon of water, works against bacteria and many viruses.
- It needs **10 minutes of wet contact**: apply it to the scrubbed surface and keep it wet for the full time rather than wiping it straight off.
- Rinse with plenty of clean water until no bleach smell remains, then let everything dry completely and air out. Plan on a full day of airing before the tortoise goes back in.

Where to do it

In a utility sink, a dedicated plastic tub, or outdoors. Never in the kitchen sink, a bathroom sink, or a bathtub people use, and never anywhere food is prepared. Wash any bedding, decor or gear separately from dishes used for people. The soaking tub is a dedicated plastic tub too, emptied into the toilet.

WHY THE RULES ARE STRICT

Healthy reptiles carry *Salmonella* bacteria in the gut and shed them in droppings without being ill, and a negative test does not change that, because shedding comes and goes. The route into people is hands and surfaces to mouth. Wash your hands with soap after every contact with the tortoise, its enclosure, its water or its substrate (page 18).

DURING QUARANTINE

A new tortoise has its own dishes, hides, tools and cleaning brushes that never cross to another tortoise's enclosure, and it is cleaned and handled last each day, after every other reptile in the house (page 21).

Diet by age & how much

A grazer of tough, fibrous plants, and a tortoise that will happily eat far more than it needs.

In the wild this animal moves through low vegetation, taking a bite here and a bite there of soft-stemmed, water-rich grasses, flowers and leaves, high in fiber and low in energy. Fiber is not filler: microbes in the gut ferment it, a major nutrient source for a tortoise. Almost every captive feeding mistake replaces that diet with something richer.

AGE	HOW MUCH AND HOW OFTEN
Hatchling and growing juvenile	As much as it will eat, every day
Nearing adult size	A limited portion every day, roughly a pile the size of its own shell
Adult	The same shell-sized daily portion as a starting point, adjusted to the tortoise in front of you
Every age	Grass hay in front of it all the time, unlimited. It is the fiber that never runs out, and the one thing never rationed

The check is the tortoise

Russian tortoises tend toward overweight in captivity, and the test is what happens when one withdraws into its shell. A tortoise that looks chubby as it pulls back, with flesh crowding the shell openings, is being fed too much: cut the amount back. That is the direction of error to expect. Underfeeding a tortoise that grazes is unusual; overfeeding one that would happily eat all day is the default.

What the diet is made of

FOOD	PLACE	NOTES
Weeds and wild greens	STAPLE	A rotation of safe weeds, dandelion and plantain first, rather than cultivated grocery greens
Grass hay	ALWAYS	Unlimited, as the background the weeds sit against
Clover	ROTATE	Higher in protein than the other weeds, so a couple of times a week rather than daily
Orange and yellow vegetables	SMALL PART	Squash, carrot and sweet potato carry carotenoids, the plant pigments a tortoise turns into vitamin A, as dark leafy greens do
Spinach, chard, beet greens	OFF THE ROTATION	High in oxalate, which binds calcium (page 15)
Fruit, any animal protein	NEVER	On the never-feed list, with the reasons

How to put it down

- **Graze first, bowl second.** Safe growing plants beat anything chopped into a dish, because grazing spreads intake across the day and keeps the fiber high.
- **Mix the food together** rather than laying items out separately. Given the choice, a tortoise eats selectively and leaves the fiber.
- **On a tile or flat stone**, so the animal is not swallowing substrate with every bite.
- **In several spots**, moved daily or every other day, so it walks and searches.

Weeds, hay & growing your own

The best food for this species grows free in an ordinary lawn.

Supermarket greens are bred for water content and mild flavor; weeds are tougher, more fibrous and more varied, and they cost nothing. Learning seven common weeds by sight is the single change most likely to improve a captive tortoise's diet. For a tortoise, a weed rotation can be the bulk of the diet, not an addition to it.

WEED	HOW TO RECOGNIZE IT, AND NOTES
Dandelion	A flat rosette of deeply toothed leaves, one bright yellow flower on each hollow, leafless stem, and milky sap. The closest thing to a default staple. Feed leaves, flowers and stem
Broadleaf and narrowleaf plantain	A low rosette with strong veins running the length of each leaf and a thin flower spike. Broadleaf has wide oval leaves, narrowleaf long narrow ones. Not the banana relative
White and red clover	Three leaflets, often with a pale chevron, and a round white or pink-red flower head. Higher in protein, so a rotating item rather than the daily bulk
Sow thistle	Soft, lobed leaves with milky sap and small dandelion-like flowers on a taller branching stem. Its spines are soft
Hawkbit	A dandelion lookalike with hairier leaves and branched stems carrying several flowers. Interchangeable with dandelion for feeding, so a mix-up carries no risk
Common mallow	Rounded, slightly lobed leaves on long stalks, low and sprawling, with small pink to purple flowers. Soft and popular, and it self-seeds
Chickweed	Small oval leaves in pairs on a thin trailing stem, tiny white star flowers, and a single line of fine hairs down the stem. Grows through cooler months

Start with dandelion and plantain, which are hard to confuse with anything dangerous, then add the rest as you learn them. Rotate two or three as a base, add clover a couple of times a week, and mix in the others as the seasons bring them.

Where not to pick

- Roadsides and anywhere near traffic, where exhaust residue, runoff and winter road salt settle.
- Dog-walking verges and park lawns: parasite contamination, and frequent herbicide on public turf.
- Anywhere sprayed with herbicide or pesticide, and the mowed edges of farmed fields, which sit in spray drift.

Growing and drying your own

A garden bed or large pots you know have never been treated solve the location problem. Dandelion, plantain, mallow and chickweed establish readily from seed, and clover is sold as ground cover. Tortoises tend to eat a plant down to the roots, so grow forage somewhere else and bring it in, or sow a tray of tortoise forage seed mix for living plants to crop. Harvest a little at a time and plantain and mallow regrow within a couple of weeks. Surplus can be air-dried on a screen out of direct sun, or in a dehydrator on its lowest setting, and stored airtight. Dried weeds are a fiber source like hay, an addition to fresh food rather than a replacement.

Fruit, protein & the never-feed list

The short page that prevents a large part of the health section. Print it for the fridge.

NEVER FEED A RUSSIAN TORTOISE

Any animal protein: meat, dog food, cat food, insects. This is a herbivore, and that is not a soft preference.

Fruit: high in phosphorus and low in calcium, the opposite of what this animal needs. It stays off the menu altogether.

Iceberg lettuce, or a poor-quality commercial food as the diet: the classic route to vitamin A deficiency.

Spinach, chard and beet greens as staples: their oxalate binds calcium in the gut and can promote bladder stones.

Any plant you cannot positively identify, and anything picked from sprayed, roadside or dog-walked ground.

Buttercup and oleander, named below.

Supplements with added phosphorus, or with added vitamin D unless a vet asked for it.

Why protein matters twice

Animal protein has no place in the diet, and it also does a slower kind of damage. Protein-rich food pushes fast growth, and rapid growth on a high-protein diet is one suspected driver of pyramiding, the raised, cone-shaped scutes that never flatten again. The condition is not completely understood, and it is not purely a feeding fault, but protein and overfeeding are the parts a keeper controls at the dish (page 28).

Two plants to know on sight

PLANT	WHY, AND HOW TO TELL
Buttercup	Grows in exactly the lawns and meadows where clover and dandelion are picked, so a careless handful can include it. Listed as toxic to dogs, cats and horses, with mouth ulcers among the signs. Its flowers are glossy and waxy where dandelion and hawkbit are matte, and its leaves are lobed like a hand rather than growing in a flat rosette
Oleander	A common hedge and border shrub, not a lawn weed, but a clipping can end up in a forage pile or an outdoor pen. One of the plants most consistently named as dangerous to animals. Keep it out of reach entirely

Toxic plant lists are built on dogs, cats and horses, and plant toxicity in reptiles is barely studied. That argues for more caution, not less: if a plant is not one you have positively identified as safe, it is not food.

Treats

There is no treat food for this species. Fruit, the usual treat for other reptiles, is on the never list above, and the variety a treat would add comes from the weed rotation instead.

Calcium, multivitamin & water

A nutrient to supply reliably, not maximally, and water that a tortoise really does drink.

Calcium runs the shell and the skeleton, and a deficiency is one of the serious problems in this species: a soft shell, shell and bone deformities, kidney damage, and worse if it goes far enough. There is also a ceiling. Wild diets for these tortoises can run a calcium to phosphorus ratio at or above 4:1, but the target in captivity sits nearer 2:1, because too much calcium causes medical problems of its own.

SUPPLEMENT	HOW
Calcium carbonate	Sprinkled lightly onto food. Calcium citrate works too. Choose one with no added phosphorus
Cuttlebone	Left in the enclosure for the tortoise to gnaw. A small, cheap calcium source that lasts a good while
Reptile multivitamin	Sprinkled lightly on food. Herbivore-appropriate, with no added phosphorus
Vitamin D3	Only if a vet asks for it. A tortoise under a working UVB tube makes its own, and D3 builds up in the body

A laying female draws heavily on her calcium, which is one more reason to keep the calcium, the cuttlebone and the UVB tube in date (page 22). Ask your reptile vet to look over what you give at each yearly exam.

Water

Tortoises drink, and the belief that they do not is a persistent one that causes real harm. Without water, urates build up and can reach dangerous levels, and chronic dehydration is the main route to the bladder stones and urate blockages on page 32. Provide water in a shallow bowl, something like a plant saucer, sunk so its lip sits level with the surrounding surface. It doubles as the soaking dish.

Soaking

A dish is an offer; a soak is a guarantee. In warm water a tortoise drinks, through the mouth and through the cloaca, then empties itself, which is why the soak is where most keepers see urates at all.

WHO	HOW OFTEN	HOW LONG	WATER
Hatchling	Daily	15 to 30 min	Warm and shallow, elbow deep
Juvenile	Twice a week	10 to 15 min	Lukewarm, no deeper than the elbows
Indoor adult	About once a week	10 to 20 min	Lukewarm, no deeper than the elbows

- **Never leave it alone.** A tortoise that tips onto its back in an inch of water cannot right itself and can drown.
- **Warm to the hand, never hot.** Fill a few degrees warm, since it cools while the tortoise sits in it. A chilled tortoise sits still and does not drink.
- **A dedicated plastic tub,** emptied into the toilet. Never the kitchen sink or a bathtub people use.
- **A tortoise that drinks hard at every soak** is living in an enclosure that is too dry. The fix is the humid hide and the substrate, not more soaks.

Growth, weight & body condition

Slow and steady is right, and the scale tells you more than the eye.

There is no growth chart worth printing for this species, because a healthy Russian tortoise grows slowly and unevenly with the seasons. A hatchling starts at about 42 to 50 mm (1.7 to 2 in) of shell and 18 to 23 g, and an adult ends up 5 to 10 in (13 to 25 cm) long. What you are watching for along the way is steady, unhurried gain.

TOO FAST

Rapid growth on a rich, high-protein diet is one suspected driver of pyramiding, which is permanent. A young tortoise is not to be pushed along.

TOO SLOW

Growth that stalls does damage as well. A young tortoise is not to be starved either; steady gain on the scale is the target.

TOO HEAVY

Flesh crowding the shell openings as it withdraws means it is being fed too much. In this species overweight is the usual direction of error.

Weighing

- **Weigh in grams** on a digital kitchen scale, at the same point in the routine each time, and write it down. Because so much of the animal is shell, the scale shows a change long before your eye does.
- **Weigh a young tortoise regularly**, weigh any new tortoise on arrival for a baseline, and weigh every one to two weeks through brumation.
- **Read the log as a trend:** steady gain in a youngster, a steady weight in a settled adult. A loss you cannot explain goes to the vet with the log.

Body condition

GOOD CONDITION	POOR CONDITION
Active and alert	Inactive or lethargic
Feels heavy for its size when lifted	Feels light for its size
Pulls head and legs into the shell when handled	Sunken or closed eyes
Clear eyes and nostrils	Any discharge from the nose or eyes
Smooth shell with no cracks, pits or soft patches	Pitting, cracks, missing scutes or a soft, flexible shell

READING THE SHELL

New growth appears at the edges of each scute, the plates covering the shell. A smooth, even shell means the growing has gone well. Scutes rising into raised cones mean pyramiding has started, and the time to correct diet, temperature and humidity is now, because what is already raised stays raised (page 28).

Handling & hygiene

An observation animal with a personality, picked up for a reason rather than for its own sake.

Russian tortoises tolerate being picked up, but they neither need it nor seek it out, and most do not enjoy it. Treat this as an animal you watch and feed rather than one you hold. Handling is for health checks, soaks, weighing and moving the tortoise, and most of the risk in those moments comes from a fall. Many tortoises do come to accept a gentle rub on the shell, and many will take food from a hand: that is the relationship to build.

Picking one up

- **Scoop from below**, a palm under the plastron (the lower shell), instead of grabbing the shell from above.
- **Then hold the shell with two hands**, so the body and the legs are both supported. Firm but gentle.
- **Work low**, sitting or kneeling over the floor or a table. A fall can do serious damage to a tortoise and can be fatal.
- **Keep it brief and purposeful**, and put it back where it can walk straight into cover.
- **Wash your hands before and after**, every time.

Salmonella, plainly

Healthy reptiles, tortoises included, carry *Salmonella* bacteria and shed them in their droppings without being ill, and a negative test does not clear an animal, because shedding comes and goes. The risk is not a bite. It is hands, water and surfaces carried to the mouth. The four-inch rule on page 26 was written for exactly this reason.

- **Wash with soap** right after touching the tortoise, its enclosure, its water, its dishes or its substrate.
- **Keep it out of the kitchen** entirely, off counters and away from anywhere food is prepared or eaten.
- **Never clean its gear** in a kitchen sink, a bathroom sink or a bathtub people use. A dedicated tub does the job.
- **No kissing** the tortoise, and no eating or drinking while handling it.

WHO	THE RULE
Children under 5	Should not handle or touch the tortoise or anything from its enclosure at all. Avoidance, not just extra care
Adults 65 and older, anyone pregnant, anyone with a weakened immune system	At higher risk of serious illness: strict handwashing, and supervision of children around the enclosure
Anyone in the household who falls ill	Diarrhea, fever and stomach cramps are the usual signs. Mention the tortoise to the doctor, even if nobody remembers touching it

Daily rhythm, courtship & living alone

What a normal day looks like, what the loud moments mean, and why one tortoise is usually the right number.

A normal day

In the wild this species is active in two short windows a day, mid-morning and late afternoon, and spends the rest of the time in its burrow. A captive tortoise that basks, grazes, walks its enclosure, digs in and then disappears for hours is keeping its own schedule. It is an unusually active animal for a tortoise: it walks the whole of whatever floor it is given and rearranges anything light enough to move. It also climbs and digs better than its shape suggests, so the enclosure is checked for escape routes as a routine.

Courtship, the loudest thing it does

A courting male circles a female and bites at her head and legs, then stretches his neck out stiffly, stares her in the eye and jerks his head up and down in quick succession. Mating comes with high-pitched squeaks from him. In the breeding season males turn on each other and fight hard.

Reading what you see

WHAT YOU SEE	WHAT IT USUALLY MEANS
Basking, then grazing, then digging in for hours	A normal day
Pacing along one wall	It can see through the wall. Solid sides fix it
Circling, biting legs, head bobbing at another tortoise	A male courting
Ramming, biting, flipping another tortoise	Males fighting, or a bully. Separate them now
One of two tortoises hiding most of the day	It is being bullied, and the stress can make it ill. Separate them
A female digging test holes, restless	Likely carrying eggs, mate or no mate (page 22)
Slowing down, digging in and eating less in autumn	The brumation drive (page 24)
Drooling or frothing on a hot day	Overheating. Shade at once, vet the same day

Living alone

- **Never two males.** Males do not tolerate each other. They fight viciously, and the loser ends up injured, flipped onto his back, or chronically stressed and prone to disease.
- **A male with a female** is hard on her: he circles and bites at her head and legs, and with nowhere to escape she takes that attention constantly. The pair will also probably produce eggs.
- **Two females** are usually more tolerant, but not always: one can become a bully. A shared enclosure needs at least 10 sq ft more floor for each added tortoise, and enough hides, plants and logs to break every sightline.
- **One is the default.** House a tortoise alone, or two females with caution.

Enrichment & common mistakes

Giving digging, grazing and exploring somewhere real to happen, and the errors that cause most trouble.

Enrichment copies the wild pattern: a burrow to disappear into, a daily round between shelter and forage, and a diet that changes constantly. It never replaces correct UVB, temperature and enclosure size, which come first.

Digging

- A deep zone of 6 to 12 in (15 to 30 cm) or more of soil mix, or a dedicated dig box.
- A partly buried hide, a terracotta half-pot or a box of loose soil gives it a start it can dig out further.

Grazing

- Scatter food in several spots instead of one bowl, and move the spots daily or every other day.
- Tuck greens into the folds of a crumpled towel or a forage mat.
- Pots of safe weeds, or a tray of tortoise forage seed mix to crop as a living plant.

Exploring

- Several hides facing different directions, plus hollow logs, cork flats, flat stones, low platforms and gentle slopes that make routes.
- Move one non-essential item every 1 to 4 weeks, depending on the tortoise. Leave the warm and cool hides where they are: they are its security.

Outdoors

- Supervised or fully secure outdoor time whenever weather and security allow is the richest enrichment there is (page 11).

The mistakes that cause most trouble

THE MISTAKE	WHAT TO DO INSTEAD
A glass aquarium instead of an open table	An open-topped table with solid walls (page 6)
Damping or misting the whole enclosure	A dry surface over a moist hide and damp depth
Substrate too shallow to dig in	At least 4 in (10 cm) for a juvenile, 6 in (15 cm) for an adult, deeper in a dig zone
A coil or compact UVB bulb, or a tube past its date	A linear T5 HO tube, replaced on the calendar (page 9)
Reading the air instead of the basking surface	An infrared gun on the surface
Grocery salad and fruit as the diet	Weeds and grass hay, no fruit
Feeding generously "to help it grow"	A steady, unhurried weight gain (page 17)
Two males together, or a second tortoise with no extra space	One tortoise, or two females with room and cover
No fecal test on a new tortoise	A fecal test on arrival, symptoms or not (page 21)
Leaving a soaking tortoise alone	Stay in the room for every soak: an upended tortoise can drown in an inch of water

Choosing a tortoise, quarantine & the first vet visit

What a healthy one looks like, and why a new tortoise lives apart for months.

Choosing

Buy captive-bred from a reputable breeder whenever possible. Wild-caught animals, and sales where nobody can say where the tortoise came from, carry a real head start on disease: imported tortoises arrive stressed and carrying parasites, and die at a higher rate. Then look at the animal itself.

LOOK FOR	WALK AWAY FROM
Active and alert	Inactive or lethargic
Feels heavy for its size	Sunken or closed eyes
Pulls its head and legs into the shell when picked up	Any discharge from the nostrils or eyes
A smooth shell, not cracked, pitted or missing scutes	Soft or infected-looking patches on the shell

The first vet visit

Book a reptile vet before or right after pickup, and have the tortoise examined within a week at the latest. Every new tortoise gets a fecal test for gut parasites, symptoms or not; roundworms and protozoa are common, especially in wild-caught animals, and often cause no signs at all. After that, an exam at least once a year with a fecal test at every visit.

Quarantine: six months, not weeks

Tortoise herpesvirus is the reason. It is shed in mouth secretions and spread mostly by ingestion, so a shared water dish, substrate or hands moving between animals can carry it. There is no cure, survivors carry it for life, and a carrier can look healthy for months. In Marenzoni and colleagues' 2018 outbreak investigation, the source tortoise had finished a full year of quarantine and showed nothing for three years beyond an occasional wet nose; when it broke, 75% of the twenty tortoises exposed died. Time alone caught nothing. Testing would have.

- **At least six months**, and six months to a year is the real range, apart from any other tortoise or turtle.
- **A separate room or airspace**, not a second enclosure in the same room.
- **Its own dishes, soak tub, hides and brushes**, which never cross to another enclosure.
- **Fed and handled last** each day, with hands washed or gloves changed between animals every time.
- **Tested, not just timed**: a full exam at the start and the end, and during it a blood test (serology) and a PCR test (polymerase chain reaction, a lab test that finds a germ's genetic material in a swab), the blood test repeated several weeks later because antibodies take time to appear.
- **A fecal screen** before the tortoise joins any other, which is also how the kidney parasite *Hexamita parva* is kept out of a group (page 31).

Sexing & females that lay

Easy to tell once grown, and a female with no male in the house can still lay eggs.

Telling males from females

FEATURE	MALE	FEMALE
Tail	Much longer, usually carried tucked to one side. The quickest cue in this species	Shorter
Vent position	Farther out toward the tail tip	Closer to the shell edge
Plastron (lower shell)	Concave, dished inward. The most useful cue on most adult tortoises	Flat, leaving room for eggs to pass
Size and shell	Slightly smaller	Slightly larger, with flared scutes

None of these are precise on a young animal, where the plastron has barely dished and the tail difference is slight, so treat an early call as provisional and check again as it grows. A vet or an experienced keeper can confirm. Sexual maturity comes at around 10 years, and a mature female can lay two or three clutches in a season. Sex matters before any pairing, because a male and a female together will probably produce eggs.

A female doesn't need a male to lay

A female builds eggs on her own cycle and still has to pass them whether or not a male was ever involved. The eggs will not hatch, but the process and its risks are the same. An unmated female is in fact the likelier one to hold a clutch, since it is usually fertilized eggs that get laid. So every female owner, not just breeders, needs a plan for where she lays.

A nest site before she needs one

A missing or unsuitable nest site is one of the causes of egg retention, and an entirely preventable one. The fix is a deep, loose area of substrate she can dig a hole her own body length into and fully bury the clutch, there all the time rather than built once she starts pacing. If the everyday substrate is too shallow, a deeper section or a separate container filled well past the normal depth works. With nowhere acceptable to dig, a female may simply hold her eggs, for up to six months, waiting for somewhere better.

Signs she is carrying eggs

- Restlessness and test-digging, often abandoned and started again over days.
- Eating less, sometimes nothing for days or weeks, while staying bright, active and alert. That last part is what makes it normal.

DO NOT PRESS ON HER BELLY

Pressing on her belly to feel for eggs can crack one inside her, and a cracked egg is a vet emergency. If you think she is carrying eggs and something seems wrong, a vet confirms with imaging instead of hand pressure.

Egg binding

Egg binding, or dystocia, is a female unable to pass her eggs. The question is the animal, not the calendar.

Telling normal laying behavior from a problem is genuinely hard in tortoises, because a female can hold eggs on purpose for weeks and look exactly like one that is stuck. Digging and straining are not the useful signal. What separates the two is whether she stays bright, active and alert between attempts.

SIGN	LIKELY A NEST-SITE PROBLEM YOU CAN FIX	LIKELY DYSTOCIA: VET THE SAME DAY
Appetite	Reduced, sometimes absent for days or weeks	Absent, and it does not come back
Alertness	Bright, active and responsive despite not eating	Weak, lethargic or unresponsive
Digging and straining	Restless test-digging that pauses and resumes over days	Straining that goes on with no progress toward laying
Abdomen	Rounder, otherwise normal	Distended, and it stays that way
Given a proper nest site	Often resolves once she has somewhere acceptable	No improvement even with a good site

Why it happens

Nonobstructive: fixable at home

No suitable nest site, the wrong temperature or humidity, a poor diet, or dehydration. These are husbandry gaps, and they can improve once the gap is fixed. Simply giving her a quiet area with the right substrate can be enough to let her lay.

Obstructive: a vet case

An oversized or misshapen egg, scarring, a misshapen pelvis, or a narrowed oviduct. These are physical blockages: no better setup fixes them, and a vet needs imaging to find them.

EGGS HELD INTO AUTUMN

In a species that brumates, eggs held into late summer or early autumn may be the body preparing for the winter slowdown rather than dystocia. A female holding eggs going into fall while still bright and eating is a different picture from one that has stopped eating and gone flat, whatever the date.

What a vet does

Imaging comes first, because the plan depends on whether there is a blockage. Without one, supportive care, fluids by vein or into the bone, calcium and vitamin injections, and a hormone such as oxytocin to make the oviduct contract, can help her pass the clutch herself. The dose is a species-specific veterinary decision, never something to estimate at home. With a real obstruction, surgery is the usual answer, since no hormone moves a physical blockage. Left untreated, prolonged retention carries serious downstream risks, and a difficult clutch can leave infection behind.

Brumation: the decision & the protocol

Built into the species, optional indoors, and dangerous done casually.

Brumation is the reptile winter slowdown: cooler temperatures and shorter days push a tortoise into weeks of near-dormancy. Russian tortoises are built for it, and wild-caught adults especially may refuse food or show stress if stopped. Indoors there is a real choice: full lighting, heat and feeding all year keeps the tortoise active instead. Both routes are defensible; a brumation drifted into without the steps below is not.

DO NOT BRUMATE A TORTOISE THAT IS

Underweight · showing any respiratory signs · sick, or recovering from illness in the last year · under about 4 years old · not vet-checked this autumn. That tortoise overwinters warm, never below 60°F (16°C), and tries again next year.

The protocol, if you say yes

STEP	WHAT TO DO
1. Vet check	Every autumn, with a fecal parasite screen. A hidden infection or parasite load becomes serious once the metabolism drops
2. The fast	Stop feeding 1 to 3 weeks before the cool-down, the longer end for a bigger tortoise, with the heat kept on at 70 to 80°F (21 to 27°C) for the whole fast so the gut empties. Food left in a cold gut rots
3. Cool down	After the fast, heat off, then down by roughly 9°F (5°C) a week into the band below
4. Hold	35 to 50°F (2 to 10°C), as close to 41°F (5°C) as you can. Below 35°F (2°C) or above 60°F (16°C) is unsafe: too cold risks blindness or death, too warm burns through the reserves it needs to wake
5. Length	A ceiling, not a target: 10 weeks at most for a small tortoise and 14 for a large one, or a shorter 6 weeks to 3 months
6. Weigh	Every 1 to 2 weeks. About 1% of body weight a month is normal; a 1,000 g tortoise losing about 10 g a month is fine. Past 7% in total, end it
7. Wake	Warm slowly: room temperature for the first few hours, then heat. Once it is alert, a warm soak of at least 30 minutes, up to twice a day. Easy, succulent food first

Where it sleeps

- **A dedicated fridge**, used for nothing else: a probe thermometer rather than the dial, the door opened about 30 seconds a day for air, and a bowl of water in the bottom for humidity.
- **Or an insulated box**: a rodent-proof, padded outer box with mesh over its air holes, and an inner box filled with polystyrene chips, dry leaves or shredded paper. No hay or straw.
- **Never unmonitored cold**. A tortoise left to find its own spot in a garden or garage faces freezing, dehydration and rodents with nobody reading a thermometer.

END IT NOW IF

Weight loss passes the action point · any discharge from the nose, eyes or vent · the temperature dropped below freezing, even briefly · any sign of illness. After waking, a tortoise that has not urinated or eaten within about a week sees a vet.

Is a Russian tortoise legal where you live?

Legal almost everywhere in the United States, with one ban people never expect.

This page covers all 50 states, the District of Columbia and New York City, which has its own rules: 52 jurisdictions. A Russian tortoise is **legal with no rule at all in 42**, **banned in 1**, **permit-only in 5**, and **legal on a condition in 4**.

WHERE	STATUS	THE RULE
Colorado	Banned	Tortoises are cleared only if native to the tropical and subtropical region, which the rule fixes between 26 degrees north and 26 degrees south. This is a Central Asian steppe animal, so it fails, becomes regulated wildlife, and the state issues no license to keep regulated wildlife as a pet.
New Jersey	Permit	New Jersey Administrative Code 7:25-4.5 requires a permit for any exotic reptile not on the exempt list, and this species is not on it. A personal possession permit, before the animal arrives.
West Virginia	Permit	West Virginia Code section 20-2-13 bars bringing live wildlife into the state, or keeping it there, without an importation permit, captive-bred or not. No application is published: ask the state in writing before the tortoise crosses the line, and keep the reply.
New Mexico	Permit	Group II on the state's importation list: an importation permit before the animal comes into the state.
Delaware	Permit	Not on the exempt list, so a \$25 Individual permit from the Department of Agriculture before anyone brings one in or keeps one.
Massachusetts	Permit	Turtles are generally exempt, but not a species the IUCN lists as threatened, and this one is Vulnerable, so a permit is always required (321 CMR, the Code of Massachusetts Regulations, section 9.01). The state says it does not issue permits for keeping a wild animal as a pet.
Hawaii	Conditional	The whole genus <i>Testudo</i> is on the List of Conditionally Approved Animals, so it may be imported and kept on the Department of Agriculture's conditions. Rare for Hawaii, where not being listed is enough to be banned.
New York City	Conditional	The city Health Code bars turtles and tortoises under 4 in (10 cm) of shell. An adult is lawful; a hatchling is not.
Vermont	Conditional	On the Unrestricted list as captive-bred only, arriving with a statement of origin or a certificate of veterinary inspection.
Minnesota	Conditional	Turtles fall under an older license rule, but no license is needed for one bought at a retail outlet.

No rule at all: 42 jurisdictions

Alabama, Alaska, Arizona, Arkansas, California, Connecticut, District of Columbia, Florida, Georgia, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Maryland, Michigan, Mississippi, Missouri, Montana, Nebraska, Nevada, New Hampshire, New York State, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, Washington, Wisconsin and Wyoming. Montana clears tortoises over 4 in of shell by statute. In eleven of these, California, Illinois, Kentucky, Maryland, Michigan, Montana, Nevada, North Carolina, Ohio, Oregon and Washington, cities or counties may be stricter, so check your local ordinance (a city or county law) too.

NOT LEGAL ADVICE

State legal status as of October 2026. This is a summary of state rules, read between August and October 2026, and not legal advice. Rules change. Before you buy or move, check with your state wildlife agency and with your city or county.

The four-inch rule & the trade

The rule that actually shapes how these tortoises are sold is federal, and it is not the one people quote.

Almost everyone has heard some version of it: turtles under four inches are illegal. That is not what the regulation says. The rule is 21 CFR 1240.62 (Title 21 of the Code of Federal Regulations, section 1240.62), a 1975 regulation of the US FDA (Food and Drug Administration) written to reduce *Salmonella* infection in small children. It says that live turtles and viable turtle eggs with a shell under 4 in (10 cm) long may not be sold, held for sale, or offered for any other kind of commercial or public distribution.

- **It is about selling, not owning.** Nothing in it stops anyone keeping a small tortoise.
- **It is about size, not species.** It catches a hatchling Russian tortoise for the same reason it catches a hatchling red-eared slider.
- **It exempts private keepers.** Subsection (c) excludes sale and distribution not in connection with a business. A private keeper rehoming a hatchling is inside the rule; a shop doing the same thing is not.

That exception is why hatchlings move hand to hand through private keepers while shops stock only animals past the four-inch line, and it is the real reason a Russian tortoise you buy is usually already a few years old. New York City and Montana write the same four-inch line into their own rules.

Why Colorado is the surprise

Colorado allows what its lists name and nothing else, and the tortoise family is on the list, so a tortoise looks cleared. But the whole reptile block carries a qualifier: native to the tropical and subtropical region, fixed at 26 degrees either side of the equator. The Russian tortoise comes from Uzbekistan, Kazakhstan, Afghanistan and Iran, far north of that line, and the state's own reptile list records it as native outside the tropics. A family-level clearance is never the end of the question.

International trade

The species is on Appendix II of CITES, where trade runs on export quotas that the countries it comes from set for themselves, and it is assessed as Vulnerable by the IUCN. It has been traded heavily since the mid-1970s, and a wild-caught supply continues alongside captive breeding. Crossing an international border with one is a CITES question as well as a state one: check the import rules before you travel.

WHAT ACTUALLY CONSTRAINS YOU

Since the law barely does, plan around what the species imposes: a digger that will tunnel out of a pen that looks secure, a real UVB requirement indoors, and a lifespan long enough that rehoming is a real possibility rather than a hypothetical.

Red flags & finding a vet

A hardy animal that hides decline for a long time, then declines quickly.

Russian tortoises are hardy, and most of their health problems trace back to a handful of husbandry basics that are within an owner's control. The pattern is worth learning as a pattern: weak UVB and too little calcium cause metabolic bone disease; cold, damp or drafty conditions cause respiratory infection; temperature, humidity and diet through the growing years decide whether pyramiding happens; and an unscreened new tortoise brings parasites and viruses in with it.

GO TO A VET NOW

Wheezing, nasal discharge, open-mouth breathing, or lethargy (respiratory infection, which does not clear on its own) · straining while making noise (a urate blockage) · watery, jelly-like urine, or straining to urinate · a crack, puncture or bleeding in the shell · soft, discolored or foul-smelling patches on the shell · a soft or deformed shell with weak limbs · a female weak, flat or straining with no progress · drooling or frothing on a hot day

Within a few days

WHAT YOU SEE	WHERE IT IS COVERED
Swollen eyelids, eyes closed, a swelling behind the eye	Vitamin A deficiency and abscesses (page 32)
Watery droppings more than once in a row	Often parasites (page 31)
Gritty, yellow or orange urates	Dehydration (page 33)
A beak that has grown long, or trouble biting	Beak overgrowth, often a sign of something else
Raised, cone-shaped scutes appearing	Pyramiding: change the husbandry now
A runny nose that comes and goes	A carrier infection is possible (page 29)

Finding a vet

Find a reptile-experienced vet before you need one, not on the day. An exotic-animal vet is meaningfully more reliable for this species than a general clinic. The Association of Reptile and Amphibian Veterinarians keeps a find-a-vet directory of its members. Write the number on the emergency card. A routine exotic-vet checkup runs around \$75; illness that needs tests or medication commonly runs \$150 to \$300 or more.

What to bring and say

- The weight log, and the weight today.
- The basking surface and cool-end readings as measured, not as set, and the age of the UVB tube.
- What it eats, in rough proportions, and any supplements.
- When it last ate, drank, and passed droppings and urates, and a fresh dropping in a sealed bag. A fresh sample matters, because some parasites are easiest to see while they are still moving.
- Whether another tortoise or reptile has joined the household.

Metabolic bone disease & pyramiding

The two conditions that are written permanently into the shell.

Metabolic bone disease

Metabolic bone disease (MBD) is among the most common conditions in pet tortoises. It comes from too little UVB or too little calcium in the diet: without UVB the tortoise cannot make the vitamin D3 it needs to absorb calcium, so however much calcium is in the food, the shell and skeleton go short.

- **Signs:** a soft or deformed shell and weak limbs.
- **What to do:** see a vet, every time. Correcting the UVB and the calcium is part of the treatment, not a substitute for it.
- **Prevention** is the whole story, and it is not optional: a linear T5 HO tube at a UVI of 3.0 to 4.0, replaced on schedule, plus a calcium source from the start, calcium lightly on food and a cuttlebone in the enclosure.

Pyramiding

Pyramiding is abnormal, raised growth of the scutes, the plates covering the shell, into cone shapes instead of a smooth curve. The condition is not completely understood. It occurs commonly in young tortoises kept at the wrong temperature and humidity, or fed an imbalance of calcium, phosphorus or protein. There is no single settled cause: low humidity during growth and the growth rate itself are both implicated. What is certain is that the growing years decide it, and that it does not reverse afterward.

SUSPECTED DRIVER	WHAT A KEEPER CONTROLS
Fast growth on rich, high-protein food	A grazing diet of weeds and hay, no animal protein, and the overweight check at withdrawal (page 13)
An enclosure dry everywhere, with no damp retreat	A dry surface over a moist hide and damp depth to dig into
Wrong temperatures	The basking and cool-end targets, measured rather than assumed
Calcium and phosphorus out of balance	Calcium with no added phosphorus, and no fruit
Chronic mild dehydration	Water always available and the soak on schedule

WHAT EXISTING PYRAMIDING MEANS

Raised scutes are permanent, so treat existing pyramiding as a record of past conditions rather than something to fix. What you can still change is the next growth, at the edges of each scute. Correct the diet, the temperatures and the humidity, and watch the new edges grow in flat.

Respiratory infection & herpesvirus

The urgent one on this list, and the one that can take a whole group of tortoises.

Respiratory infection

Wheezing, nasal discharge, open-mouth breathing and lethargy are the signs. The causes are temperatures running chronically low, too much humidity across the enclosure, and drafts, which is why ventilation and a dry surface are health measures rather than comfort ones. Vitamin A deficiency can bring it on too.

IT DOES NOT CLEAR ON ITS OWN

A respiratory infection in a tortoise needs a vet and antibiotics, and it does not wait. While you arrange care, raise the temperature and make sure the tortoise is drinking. Do not raise the humidity, which is the common and wrong instinct.

Herpesvirus

Tortoise herpesvirus is confirmed in Russian tortoises and in many other true tortoises. It spreads mostly by mouth, through shared water, substrate and hands, and there is no cure: treatment only supports the animal and controls the infections that follow. Survivors carry the virus for life and shed it on and off, most of all after brumation.

LOOK FOR	NOTES
Nose and eye discharge	A mild case can look like nothing more than a touch of eye irritation
Plaques on the tongue and roof of the mouth	Often with drooling
Trouble eating or swallowing, regurgitation	With lethargy
Pneumonia or neurological signs	In severe cases

Because a carrier can look entirely well, watching for symptoms is not screening. That is why every new tortoise does a tested quarantine of at least six months (page 21), and why a new tortoise never goes in beside an established one of another species either.

Mycoplasma, the runny nose that keeps coming back

Mycoplasma agassizii is the bacterium behind upper respiratory tract disease in tortoises, and it has the same carrier problem: it has been found in captive tortoises with no symptoms at all. A runny nose that comes and goes is never just a cold in a tortoise. It is a reason for a vet visit and for keeping that animal apart from any other.

Shell rot, shell trauma & abscesses

The shell is living bone, and it gets infected and injured like any other tissue.

Shell rot

Shell rot shows as soft, discolored or foul-smelling patches on the shell. It is a bacterial or fungal infection that usually gets in through a cut or a scrape, and an overly damp enclosure often makes it worse, one more reason to keep the surface dry. Untreated it can spread into the bloodstream as septicemia, which is life-threatening. It is a vet case from the first patch.

Shell trauma

A shell has its own blood and nerve supply, so a crack or puncture is a wound, not cosmetic damage. The commonest cause is the household dog: chewing a tortoise like a toy leaves punctures or breaks whole sections away, sometimes exposing the organs underneath. Another tortoise's bites land on the neck and feet.

- **Same-day vet** for any crack, puncture or bleeding. A fresh fracture in a well-fed tortoise can usually be repaired; waiting invites bone infection.
- **Never leave a tortoise and a dog together unsupervised**, whatever the dog's record, in the house or the yard.
- **Falls count too.** Handle low, sitting or kneeling, because a drop can be fatal.

Abscesses

A tortoise abscess is a hard, tumor-like lump rather than a soft pocket, because the pus inside is thick and dry, close to cottage cheese. The classic sites are the ear, which shows as a large swelling on the side of the head just behind the eye, and the mouth, including the roof of it. Many trace back to vitamin A deficiency (page 32).

THEY DO NOT DRAIN ON THEIR OWN

A vet opens and cleans out the abscess, flushes the site, cultures the bacteria to choose the right drug, and follows with antibiotics. Waiting does not shrink it.

Parasites & *Hexamita*

Common, mostly invisible, and the reason every new tortoise is screened.

Gut parasites

Roundworms and protozoa (single-celled parasites) are fairly common in tortoises, especially wild-caught ones, and often cause no signs at all. A fecal test is worth doing for every newly acquired tortoise, symptoms or not, and at every yearly exam after that. Repeated watery droppings are a classic sign of protozoal or parasitic disease and a reason to test sooner.

Hexamita parva: the kidney parasite

Hexamita parva is a flagellate, a single-celled protozoan that swims with whip-like tails, not a worm, and it attacks the kidneys and urinary tract rather than the gut. Russian tortoises and their relatives in the genus *Testudo* are notably affected. Hallinger and colleagues' 2018 study of pet tortoises in Germany found it invading the kidney tubules, and described long-running infection as able to cause severe kidney inflammation, tissue death and gout, occasionally fatally.

STAGE	WHAT TO KNOW
Signs	Abnormally watery, jelly-like urine with soft droppings: the kidneys are already under strain. By the time the tortoise looks unwell, the disease is often well established
Diagnosis	A fresh urine or fecal sample under a microscope. Fresh, because the parasite moves and is easiest to spot before it dies off in an older sample
Spread	Through cysts shed by an infected tortoise, so it moves easily through a shared enclosure or water dish
Treatment	Metronidazole is the usual drug. The vet sets the dose and course for your tortoise's weight, kidney function and stage of infection. Never a general figure from home

WATERY OR JELLY-LIKE URINE

Is worth a vet visit on its own. Isolate the tortoise from any others while you wait for results, and screen any tortoise with a fecal test before it joins another or a new enclosure.

Vitamin A, bladder stones & beak overgrowth

Three conditions from three corners of husbandry: the diet, water, and the texture of food.

Vitamin A deficiency

A diet built on iceberg lettuce or a poor-quality commercial food leaves a tortoise short of vitamin A, and the shortage changes the skin and the glands that make mucus. The signs are easy to blame on something else: poor appetite, lethargy, swollen eyelids often with a pus-like discharge, a swelling at the ear from an abscess, respiratory infection, and kidney failure late on. A vet treats it with vitamin A by mouth or injection while the diet is fixed. Dark leafy greens, weeds and orange vegetables carry the carotenoids a tortoise turns into vitamin A, which is why a weed-based diet prevents it.

Urate blockages and bladder stones

A cloacolith is a stone of urates, the crystalline part of the urine, lodged in the cloaca, the single opening that waste and eggs leave by. A large one blocks it. A tortoise with one strains to pass a bowel movement and makes noise as it strains, which is a true emergency: a vet removes it under sedation with pain relief, breaking it down and flushing it out. A soak will not shift a stone that is already wedged.

Bladder stones are common in tortoises and are made almost entirely of urate salts. In Takami and colleagues' 2021 survey of 101 tortoises at three Japanese veterinary hospitals, Russian tortoises were 8.9% of the cases, and every stone analyzed was ammonium acid urate. Chronic dehydration is the mechanism most often pointed to: a tortoise's kidneys cannot concentrate urine, so a thirsty or overheated one pulls water back out of the bladder and the urate crystallizes. The same survey could not tie stones to diet. Straining and a poor appetite were the commonest signs, some tortoises showed none until the stone was large, X-ray finds it, and surgery removes it.

Prevention for both is water: a dish always available, the soak on schedule, and a moist hide.

Beak overgrowth

A tortoise's beak grows all the time and wears down on tough food. An overgrown one can come from metabolic bone disease, a misaligned jaw, a diet of soft foods, or an old injury, so a beak that keeps overgrowing is a symptom to investigate rather than a grooming job. Since soft food is one of the causes, the coarse weeds and hay of a proper diet are the prevention.

TRIMMING IS A VET'S JOB

The beak has its own blood and nerve supply. A trim done wrong causes severe pain, bleeding and lasting disfigurement.

Reading droppings, urates & hydration

A free health check every day, and the soak is where you get to see it.

A tortoise passes two things together through its single opening: the solid dropping, and urates, the white to cream paste a reptile makes instead of liquid urine. Urates are the best hydration gauge an owner has. Many owners never see them on a normal day, because they land in the substrate and get buried, which is why the soak, when most tortoises drink and empty themselves, is the time to look.

WHAT	NORMAL	ACT ON IT
Droppings	Fibrous and fairly bulky, closer to a rabbit's or a horse's than to a lizard's	Watery or unformed more than once in a row: parasites are a classic cause (page 31)
Urates	Smooth, creamy, white to cream	Gritty, granular, dry, yellow or orange: dehydration at least, and a road toward kidney trouble and gout if it continues
Urine	Passed with the urates, often during a soak	Watery and jelly-like, with soft droppings: a <i>Hexamita</i> sign, vet visit
Passing	Easy	Straining, especially with noise: a urate blockage, vet now
Blood or smell	None, and the usual smell	Bright red or dark, tarry blood, or an unusually strong smell
How often	Every few days to about once a week for an adult; more often for a youngster	A real change from your own tortoise's usual rhythm, in either direction

Gritty urates are the sign to catch

Chronically gritty, granular, off-white urate is clear evidence of extended dehydration, and it comes before the urate blockage and the bladder stones on page 32. When a soak turns it up, review the water dish and the soak schedule now rather than waiting to see.

Hydration, in four routines

- **A water dish**, sunk level with the substrate, washed and refilled daily.
- **The soak on schedule**: hatchlings daily, juveniles twice a week for 10 to 15 minutes, adults about weekly for 10 to 20 minutes, lukewarm and no deeper than the elbows, never unattended.
- **A moist hide** and damp depth in the substrate, which is the real fix for a tortoise that drinks hard at every soak.
- **Fresh, water-rich food**: weeds and greens carry moisture that hay alone does not.

WHAT A SOAK SHOULD PRODUCE

A soaking tortoise drinks, urinates and defecates. That is the point of it. If the tub is still clean when the soak ends, the animal had nothing to give, and that is the reading to act on.

Setup checklist & targets

Essential setup checklist

- ☐ Open-topped table, 4×2 ft (120×60 cm) at least, 7×3.5 ft (210×105 cm) the target (page 6)
- ☐ Solid walls a foot (30 cm) high, nothing climbable against them
- ☐ Halogen basking bulb on a dimming thermostat, probe on a clip or suction cup (page 8)
- ☐ Every lamp at least 4 to 6 in (10 to 15 cm) above the highest point the tortoise can reach
- ☐ Linear T5 HO UVB tube, 10 or 12% rated, a third to half the table length, at its mounting distance (page 9)
- ☐ Timer for every light
- ☐ Infrared temperature gun, digital probe thermometer and hygrometer
- ☐ Topsoil and coir mix, 4 in (10 cm) deep for a juvenile, 6 in (15 cm) for an adult, deeper dig zone (page 10)
- ☐ Warm hide, cool moist hide lined with damp peat moss
- ☐ Shallow water dish sunk level, and a feeding tile
- ☐ Grass hay, weeds, calcium with no added phosphorus, multivitamin, cuttlebone (page 16)
- ☐ 6500 K daylight lamp alongside the UVB tube
- ☐ Digital scale reading in grams
- ☐ A plastic tub for soaks and cleaning gear, never the kitchen sink (page 12)
- ☐ Reptile vet found, first exam booked within a week (page 21)

Targets

TARGET	°F	°C
Basking surface	About 95	About 35
Daytime air (the cool-end probe, moved to mid-table for a spot check)	75 to 85	24 to 29
Cool end	the 70s	21 to 26
Night: fine down to / act below	60 / 60	16 / 16
Humidity at the surface	40 to 50% adult; 40 to 75% juvenile; moist hide always	
UVI at the basking area	3.0 to 4.0	
UVB tube distance to the tortoise's back	14 to 16 in (35 to 40 cm) through mesh; 17 to 20 in (43 to 50 cm) with no mesh	
Light cycle	12 hours on, 12 off, or about 14 in summer to 10 in winter	
Soak	Hatchling daily; juvenile twice a week, 10 to 15 min; adult weekly, 10 to 20 min	

Before your tortoise comes home

- ☐ Table set up and running for a few days, readings confirmed with the gun and the thermometer
- ☐ Thermostat tested: watch it dim the bulb as the probe warms
- ☐ UVB tube date written on the equipment log (page 44)
- ☐ A separate quarantine room ready if another tortoise lives here

Emergency & quick targets

CALL A REPTILE VET NOW IF

Wheezing, nasal discharge, open-mouth breathing or lethargy · straining while making noise · watery, jelly-like urine or straining to urinate · any crack, puncture or bleeding in the shell · soft, discolored or foul-smelling shell patches · a soft or deformed shell with weak limbs · a female weak, flat or straining with no progress · drooling or frothing on a hot day · discharge or a freeze during brumation, or no urine or food within a week of waking

READING	°F	°C
Basking surface	About 95	About 35
Cool end	the 70s	21 to 26
Floor, indoors and out (act if below it)	60	16
Brumation band, if chosen	35 to 50, ideal 41	2 to 10, ideal 5
Humidity	40 to 50% at the surface (juvenile 40 to 75%), moist hide always	
UVI at the basking area	3.0 to 4.0; T5 HO tube replaced every 6 to 12 months by model	
Soak	Adult weekly, 10 to 20 min; juvenile twice a week, 10 to 15 min; elbow deep, never alone	

Your numbers

Reptile vet _____

UVB tube fitted _____

24-hour exotic emergency clinic _____

Replace UVB by _____

Tortoise's name and hatch date _____

Last vet visit and fecal test _____

Baseline weight (g) _____

Long-term caregiver _____

HAVE THIS READY WHEN YOU CALL

The weight log and today's weight, the basking and cool-end readings as measured, the UVB tube's age, the diet, when it last ate, drank and passed anything, and a fresh dropping in a sealed bag. Transport in a ventilated tub with a towel to grip, kept warm (page 41).

Budget: setup & shopping list

United States retail ranges, last checked October 2026 and rounded to the nearest \$5. Prices move; treat this as a snapshot, not a quote.

ONE-TIME SETUP	COST
Tortoise table, bought or built, 4×2 ft or larger; check the dimensions before you buy	\$125 to \$250
T5 HO UVB fixture about a third to half the table length (check the length against your enclosure), 10 or 12% tube included	\$60 to \$130
Halogen basking bulb	\$15 to \$25
Dome fixture for the basking bulb	\$10 to \$20
Dimming thermostat for the basking bulb	\$20 to \$35
Plug-in timer for the lights	\$10 to \$20
Infrared temperature gun for the basking surface	\$15 to \$30
Digital probe thermometer and hygrometer	\$10 to \$20
Substrate, a 50/50 topsoil and coconut coir mix, 6 in (15 cm) deep for an adult; check the quantity against your table's floor	\$30 to \$70
Two hides, one warm and dry and one cool and moist lined with damp peat moss, plus cork flats	\$30 to \$70
Dig box: a storage tub of deeper substrate for the 6 to 12 in (15 to 30 cm) dig zone	\$5 to \$15
Slate tiles for a feeding tile and flat stones	\$35 to \$55
Shallow water dish big enough to soak in, sunk level with the surface	\$15 to \$30
Digital kitchen scale in grams	\$5 to \$20
First supply of plain calcium without D3, a herbivore multivitamin with no added phosphorus, and a cuttlebone	\$15 to \$30
First bag of grass hay	\$10 to \$15
Equipment, every row added up	\$410 to \$835
The tortoise, captive-bred: young \$50 to \$150, older \$225 to \$400	\$50 to \$400
First checkup with an exotic-animal vet	about \$75
To get started	about \$535 to \$1,310

The dimming thermostat, infrared gun and kitchen scale are what first setups leave out, and what prevents burns and catches weight loss. A UVB tube's wattage is not its length: a slim high-output tube of 24 watts is about 22 in (56 cm) long, 39 watts about 34 in (86 cm), 54 watts about 46 in (117 cm). Match the length in inches to a third to half of the table length.

Budget: yearly costs

What a Russian tortoise costs to keep each year once the setup on the previous page is in place, on the same rounding.

ONGOING, PER YEAR	COST
Greens, weeds and grass hay	\$140 to \$280
Supplement top-ups: calcium, multivitamin and a fresh cuttlebone	\$20 to \$35
Substrate replacement	\$40 to \$90
UVB tube, replaced yearly	\$45 to \$90
Electricity for heat and light	\$60 to \$120
Yearly wellness exam with a fecal test	\$60 to \$100
Per year, every row added up	\$365 to \$715
Which is, per month	about \$30 to \$60

THE FORTY-YEAR NUMBER, AND THE SICK VISITS

At \$365 to \$715 a year, forty years comes to \$14,600 to \$28,600 before a single illness. Illness that needs tests or medication commonly runs \$150 to \$300 or more each time.

WHERE NOT TO CUT CORNERS

The UVB tube and the dimming thermostat. Replace the tube on the calendar, not when it looks dim, and never run the basking bulb without the thermostat. Electricity rises in a cold winter.

First 30 days

Before it arrives

- ☐ Table running for a few days, every reading confirmed (page 34)
- ☐ Reptile vet found and the first exam booked for within a week (page 27)
- ☐ If another tortoise or reptile lives here, the separate quarantine room set up with its own tools (page 21)

Day 1

- ☐ Weigh in grams for a baseline, and photograph the shell top and bottom (page 43)
- ☐ Check the basking surface again with the tortoise in the enclosure
- ☐ Fill in the emergency card (page 35)
- ☐ Hands off except to move it, soak it and weigh it (page 18)

Week 1

- ☐ First exam with a fresh dropping for the fecal test, symptoms or not
- ☐ Start the soak schedule for its age (page 16)
- ☐ Watch the first droppings and urates (page 33)

Weeks 2 to 4

- ☐ Food scattered in a few spots on a tile, hay always there (page 13)
- ☐ Learn dandelion and plantain, or sow a forage tray (page 14)
- ☐ Act on the fecal result if parasites showed up (page 31)
- ☐ Check the overweight sign at withdrawal (page 17)
- ☐ Write the UVB replacement date in the log (page 44)

Days 31 to 180

- ☐ Keep quarantine running for the full six months, with the testing the vet advises
- ☐ First supervised outdoor time when weather and security allow (page 11)
- ☐ Fill in the pet-sitter sheet before you need it (page 42)
- ☐ Name a long-term caregiver and write it on the card (page 41)
- ☐ If it is a female, a nest site in place before she needs it (page 22)

WHAT SETTLED LOOKS LIKE

Out basking and walking its enclosure in the morning, grazing, digging in or disappearing for hours, and passing fibrous droppings with smooth white urates, most visibly at the soak. A tortoise that never comes out, refuses food for weeks, or passes nothing is not settling in: it needs the vet.

Symptom quick reference

SYMPTOM	LIKELY CAUSE	ACTION
Wheezing, nasal discharge, open-mouth breathing	Respiratory infection	Vet now; warmer, not damper (page 29)
Plaques in the mouth, drooling, trouble swallowing	Herpesvirus	Vet now, isolate from other tortoises (page 21)
Straining while making noise	Urate blockage (cloacolith)	Vet now (page 32)
Watery, jelly-like urine with soft droppings	<i>Hexamita parva</i>	Vet, isolate, fresh sample (page 31)
Crack, puncture or bleeding in the shell	Dog, fall, crush	Vet the same day (page 30)
Soft, discolored, foul-smelling shell patches	Shell rot	Vet; dry the enclosure surface (page 30)
Soft or deformed shell, weak limbs	Metabolic bone disease	Vet; check the UVB tube and calcium (page 28)
Swollen eyelids, eyes closed, discharge	Vitamin A deficiency	Vet; fix the diet (page 32)
Hard lump behind the eye or in the mouth	Abscess	Vet; it will not drain on its own (page 30)
Overgrown beak, trouble biting	Soft diet, MBD, jaw problem	Vet trim; coarser food (page 14)
Gritty, yellow or orange urates	Dehydration	Soak, check water and humid hide (page 33)
Watery droppings more than once	Parasites, diet	Vet with a fresh sample (page 31)
Raised, cone-shaped scutes appearing	Pyramiding	Fix diet, heat and humidity now (page 28)
Flesh crowding the shell openings	Overfeeding	Smaller portions (page 13)
Female digging test holes, eating less, bright	Carrying eggs	Nest site now (page 22)
Female weak, flat, straining with no progress	Egg binding	Vet the same day (page 23)
Slowing, digging in, eating less in autumn	The brumation drive	Decide, do not drift (page 24)
Pacing along a wall	It can see through it	Solid walls (page 6)
Ramming, biting, flipping another tortoise	Males fighting, or a bully	Separate now (page 19)
Drooling or frothing on a hot day	Overheating	Shade at once, vet the same day (page 11)
Drinking hard at every soak	Enclosure too dry	Moist hide and damp depth (page 16)

Daily, weekly & seasonal routine

Daily

- ☐ Fresh weeds and greens, mixed and scattered on the tile; hay topped up (page 13)
- ☐ Water dish emptied, washed and refilled
- ☐ Droppings, urates and uneaten fresh food picked out (page 12)
- ☐ Basking surface read with the gun, cool end with the thermometer (page 7)
- ☐ A proper look at the tortoise: eyes, nose, gait, and whether it basked
- ☐ Hands washed after every contact

Weekly

- ☐ Soak, twice for a juvenile, daily for a hatchling, and read the urates (page 33)
- ☐ Weigh and log it (page 43)
- ☐ Wipe dishes, hides and the tile; dampen the moist hide
- ☐ Walk the perimeter of any outdoor pen for fresh digging (page 11)

Every 1 to 4 weeks

- ☐ Move one non-essential item; leave the hides alone (page 20)

Every 3 to 6 months

- ☐ All-new substrate, with everything scrubbed and disinfected

Seasonal

- ☐ Spring: weeds return; outdoor time begins as the weather allows
- ☐ Summer: check the enclosure and pen on the hottest afternoons; shade fixed
- ☐ Shift the timer toward 14 hours in summer and 10 in winter, if you use the seasonal cycle (page 9)
- ☐ Autumn: the brumation decision, made on purpose, with the vet check first (page 24)
- ☐ Before nights fall toward 60°F (16°C), outdoor tortoises come in

Yearly

- ☐ Wellness exam with a fecal test (page 21)
- ☐ UVB tube replaced on its date, 12 months at the latest (page 44)

Power outages, heat waves, travel & transport

A tortoise depends on an outlet for the one thing it cannot make: body heat.

When the power goes out

A healthy adult tolerates one cool night, and this species tolerates 60°F (16°C). The risk is duration: a day or two below 60°F outside brumation leaves a reptile open to infection and off its food.

Do

- Keep a battery or wind-up thermometer in the enclosure.
- Drape thick towels over the table, leaving a vent gap.
- Move the tortoise in a ventilated tub to the warmest room.
- A wrapped hot water bottle or hand warmer beside it, never touching it.
- Offer water and a few greens only, until heat is reliably back.
- Past a day or two below 60°F, board it with a vet or a friend with power.

Don't

- No candles, gas stoves or open flame near the enclosure.
- No heat pack against the skin: a cold tortoise cannot move away in time.
- No hot water for a cold, unresponsive tortoise. Lukewarm only, once it is alert.

OUTAGE KIT

Hand warmers or a hot water bottle, two towels, a battery thermometer, a ventilated tub, vet numbers on paper.

In a heat wave

Read the enclosure and any pen on the hottest afternoon, keep the table out of direct sun, and make outdoor shade fixed. A tortoise drooling or frothing goes into shade at once and to a vet the same day.

Leaving town

TRIP LENGTH	HEALTHY ADULT	HATCHLING, JUVENILE, SICK OR GRAVID
1 to 2 nights	Often fine alone: lights on timers, thermostat on, water full, hay and greens down	A sitter once a day
3 to 7 nights	A sitter every 1 to 2 days for water, food, a dropping check and a temperature check	A sitter daily, with food portioned and instructions written
Over a week	A daily sitter who has done a practice visit, or boarding with a reptile-experienced vet. Never a general dog-and-cat kennel	

Transport and vet trips

A ventilated tub with a towel to grip, in a dark bag, no loose substrate or water dish. Warm the car first; below 65°F (18°C) on a trip over 30 minutes, a wrapped hand warmer beside the tub. Check it every 20 to 30 minutes at a stop, and never leave it in a parked car. On a move, the new enclosure runs before the tortoise arrives.

THE FORTY-YEAR VERSION OF THIS PAGE

A tortoise outlives plans. Name a guardian who has agreed, and a second behind them, and contact a tortoise rescue in advance. A pet cannot inherit money, since the law treats it as property, but a pet trust can fund its care, and all 50 states and the District of Columbia have pet trust laws: a question for an estate attorney. Leave a care file with the animal, and rehome through a rescue, never a free listing.

FILL IN AND LEAVE BY THE ENCLOSURE

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Tortoise's name _____

Normal weight (g) _____

My phone _____

Reptile vet _____

24-hour emergency clinic _____

Backup contact _____

Each visit

- ☐ Read the cool-end thermometer: the 70s°F (21 to 26°C) by day. If shown how, aim the gun at the basking spot: about 95°F (35°C). They are meant to read differently. Lights on between ____ and ____.
- ☐ Fresh greens from the labeled box in the fridge, scattered on the tile in a couple of spots. Hay topped up.
- ☐ Calcium on the food today: yes / no
- ☐ Empty, wash and refill the water dish.
- ☐ Pick out droppings with a paper towel. Normal is fibrous with a white, creamy part.
- ☐ Wash your hands before and after. Nothing rinsed in the kitchen sink.

Please don't

- Change any dial, timer or thermostat, or move the lamps.
- Take the tortoise out or outside, unless we agreed on it, or let a dog or cat near it.
- Feed fruit, meat, pet food, iceberg lettuce, or any plant from the garden.
- Soak it unless we arranged it, and then never leave the room while it is in the water.

CALL ME IF

The lights are off during the day, or the cool end reads below 60°F (16°C) · it breathes with its mouth open, wheezes, or has bubbles at the nose · it has not moved on two visits · there is a crack or blood on the shell · it strains or squeaks while passing waste · food is untouched on two visits · anything else looks wrong. If I don't answer, call the vet number above.

Owner log

So much of a tortoise is shell that the scale shows a change long before the eye does, and a written trend turns "it looks a bit smaller" into a number a vet can use. Weigh at the same point in the routine each time, ideally the soak, and look at the column as a trend.

Equipment & vet log

Equipment log

ITEM	FITTED	REPLACE OR CHECK BY	DONE
UVB tube (6 to 12 months by model)			
Basking bulb			
Thermostat and probe check			
Thermometer and hygrometer batteries			
UVI reading at the basking area			
Substrate fully replaced (3 to 6 months)			

Twelve-month planner

Tick the box in the month you did it. Month 1 is whichever month you start in.

TASK, BY MONTH	1	2	3	4	5	6	7	8	9	10	11	12
Thermostat check	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Substrate replaced				☐				☐				☐
UVB tube replaced, by model						☐						☐
Vet exam and fecal test	☐											☐
Brumation decision and vet check	Autumn, every year: _____											

Vet visit log

DATE	WEIGHT (G)	REASON AND FINDINGS	FOLLOW-UP

Enrichment checklist & log

Setup

- ☐ A dig zone 6 to 12 in (15 to 30 cm) deep, or a dig box (page 10)
- ☐ A partly buried hide it can dig out further
- ☐ Hides facing different directions, warm and cool
- ☐ Logs, cork flats, flat stones and a gentle slope, away from the walls
- ☐ A forage tray or pots of safe weeds (page 14)
- ☐ A nest area for an adult female (page 22)

Each month

- ☐ Food scattered in new spots, daily or every other day
- ☐ Greens tucked into a crumpled towel or forage mat
- ☐ One non-essential item moved; hides left in place (page 20)
- ☐ Supervised or secure outdoor time, weather allowing (page 11)
- ☐ Dig zone topped up and kept damp at depth

Enrichment log

DATE	ACTIVITY	HOW IT WENT, AND WHETHER TO REPEAT

ENRICHMENT COMES SECOND

None of this replaces correct UVB, temperature and enclosure size, which come first. Enrichment is giving digging, grazing and exploring somewhere real to happen once those are right.

Glossary, A to M

21 CFR 1240.62	Title 21 of the Code of Federal Regulations (CFR), section 1240.62: the federal four-inch rule. It restricts selling turtles under 4 in of shell, not owning them, and exempts private keepers. Page 26.
Abscess	A pocket of infection. In a tortoise it is a hard lump of thick, dry pus, classically at the ear behind the eye or in the mouth. A vet case. Page 30.
Brumation	A reptile's winter slowdown into near-dormancy. Optional indoors for this species, held at 35 to 50°F (2 to 10°C) for no more than 10 weeks in a small tortoise and 14 in a large one. Page 24.
Captive-bred	Hatched from parents kept by people rather than taken from the wild. The better buy: wild-caught tortoises arrive stressed and carrying parasites. Page 5.
Carotenoids	Orange and yellow plant pigments a tortoise turns into vitamin A, found in dark leafy greens, weeds, squash, carrot and sweet potato. Page 13.
Ceramic heater	A screw-in heat source that gives off no light, so it can run at night when a room will not hold 60°F (16°C). Page 7.
CITES	The Convention on International Trade in Endangered Species, the treaty that controls moving listed animals across borders. This species is on its Appendix II, traded on export quotas. Page 26.
Cloaca, cloacolith	The cloaca is the single opening that droppings, urates and eggs leave by. A cloacolith is a stone of urates lodged in it: straining with noise, and an emergency. Page 32.
CFR, CMR	The Code of Federal Regulations (CFR), the official book of United States federal rules, and the Code of Massachusetts Regulations (CMR), that state's equivalent. Pages 25, 26.
Dimming thermostat	A controller that lowers the power to a heat bulb instead of switching it off and on. The right kind for a halogen basking bulb. Page 8.
Dystocia	Egg binding: a female unable to pass her eggs. Told apart from normal laying by whether she stays bright and alert. Page 23.
FDA	The Food and Drug Administration, the United States agency behind the four-inch rule. Page 26.
Fecal test	A vet's microscope check of a fresh stool sample for parasites. On arrival, symptoms or not, and at every yearly exam. Page 31.
Herpesvirus	Tortoise herpesvirus: contagious, incurable, carried for life by survivors, and able to hide in a healthy-looking tortoise for months. The reason for a six-month tested quarantine. Page 29.
<i>Hexamita parva</i>	A single-celled flagellate parasite of the kidneys and urinary tract that notably affects this genus. Watery, jelly-like urine is the sign. Page 31.
Infrared temperature gun	A handheld thermometer that reads a surface from a distance. The way to check the basking spot. Page 7.
IUCN	The International Union for Conservation of Nature, which keeps the Red List of threatened species. It assesses this tortoise as Vulnerable. Page 4.
K (kelvin)	The scale for the color of light. A 6500 K lamp gives white daylight. Page 9.
MBD	Metabolic bone disease: a soft, deformed shell and weak limbs from too little UVB or calcium. Among the commonest conditions in pet tortoises. Page 28.
Moist hide	A hide lined with damp peat moss on the cool side, giving a humid retreat in a dry enclosure. Page 10.
<i>Mycoplasma</i>	A bacterium behind upper respiratory disease in tortoises, found even in tortoises with no symptoms. Page 29.

Glossary, N to Z

Oxalate	A compound in spinach, chard and beet greens that binds calcium in the gut and can promote bladder stones. Why those greens stay off the rotation. Page 15.
PCR	Polymerase chain reaction: a lab test that finds a germ's genetic material in a swab, used to screen a quarantined tortoise for herpesvirus. Page 21.
Plastron	The lower shell. Concave in an adult male, flat in a female. Page 22.
Probe	The thermostat's sensor, placed where the tortoise basks and held with a probe clip or suction cup, never tape. Page 8.
Protozoa	Tiny single-celled life forms. Some are gut parasites common in tortoises, and <i>Hexamita</i> is one that attacks the kidneys. Page 31.
Pulse proportional thermostat	A controller that sends short pulses of power to hold a steady temperature. Suits a ceramic heater. Page 8.
Pyramiding	Scutes that rise into raised cones as the shell grows. Not completely understood, linked to temperature, humidity and diet during growth, and permanent. Page 28.
Quarantine	Keeping a new tortoise fully apart, in a separate room with its own equipment, for at least six months with vet testing. Page 21.
<i>Salmonella</i>	Bacteria that healthy reptiles carry in the gut and shed in droppings without being ill. Spread hands to mouth. Page 18.
Scute	One of the plates covering the shell. New growth appears at each scute's edge. Page 17.
Serology	A blood test for antibodies, the body's response to an infection. Repeated weeks apart in quarantine, because antibodies take time to appear. Page 21.
Shell rot	A bacterial or fungal shell infection: soft, discolored or foul-smelling patches, often worse in a damp enclosure. Page 30.
T5 HO	A fluorescent tube five-eighths of an inch across (T5) in its high-output (HO) version. The UVB source for this species; coil and compact bulbs are not equivalent. Page 9.
Tortoise table	An open-topped enclosure with solid walls, preferred over a glass aquarium for ventilation and because the tortoise cannot see through it. Page 6.
Units	°F, degrees Fahrenheit, and °C, degrees Celsius, for temperature; in (inches), ft (feet), sq ft (square feet), m (meters), cm (centimeters) and mm (millimeters) for size; lb (pounds) and g (grams) for weight; min (minutes); % (percent) for humidity, UVB tube strength and weight loss.
Urates	The white to cream paste a reptile passes instead of liquid urine. Smooth and creamy means hydrated; gritty or yellow means dry. Page 33.
UVB	Ultraviolet B: the part of sunlight a tortoise needs on its skin to make vitamin D3 and absorb calcium. Window glass blocks it, and tubes lose it with age. Page 9.
UVI	Ultraviolet index: the strength of the UVB at one spot, read with a handheld meter. 3.0 to 4.0 at the basking area for this species. Page 9.
VCA	Veterinary Centers of America, now VCA Animal Hospitals: an animal hospital chain whose pet care articles are among this package's sources.
Vitamin A deficiency	A shortage from an iceberg lettuce or poor commercial diet: swollen eyelids, ear abscesses, respiratory infection. Page 32.
Vitamin D3	The vitamin a tortoise makes in its skin under UVB and needs to absorb calcium. It builds up in the body, so a supplement is only given when a vet asks. Page 16.

REFERENCE

Sources

Where the numbers in this package come from, so you can check them yourself.

Veterinary references

- VCA Animal Hospitals (Veterinary Centers of America): Common Diseases of Tortoises, Owning a Tortoise, Housing Box Turtles, and Reptiles: Dystocia. The diseases, the healthy-tortoise checks and the first exam.
- Merck Veterinary Manual: management and husbandry, nutrition in tortoises, parasitic and viral diseases, routine health care, clinical procedures, and cleaning and disinfection. Sunlight hours and the bleach dilution.
- LafeberVet: the Russian tortoise and Mediterranean tortoise information sheets, and UVB lighting for reptiles. Sexing, maturity and the calcium ratio.
- Royal Veterinary College: tortoise hibernation guidelines. Texas Veterinary Medical Foundation: brumation in turtles and tortoises. The whole brumation protocol.
- Colorado Exotic Animal Hospital: basic care of the Russian tortoise. Margaret A. Wissman, veterinarian, in Reptiles Magazine: trimming tortoise beaks. European Association of Zoo and Wildlife Veterinarians: herpesvirus infection in tortoises. Association of Reptile and Amphibian Veterinarians: the find-a-vet directory.

Research

- Marenzoni and colleagues (2018), PLOS ONE: an outbreak of testudinid herpesvirus 3 in a captive breeding facility. The year of quarantine and the 75% loss.
- Hallinger and colleagues (2018), Parasites & Vectors: parasites of pet tortoises in Germany. Hatt (2010), World Small Animal Veterinary Association World Congress Proceedings: selected endoparasites in reptiles. The *Hexamita* findings.
- Takami and colleagues (2021), Journal of Veterinary Medical Science: tortoises with urolithiasis in Japan. Lubian and colleagues (2025), Animals, and Morel and colleagues (2025), Veterinary Sciences: reproductive disorders and dystocia.
- Segura and colleagues (2020), Scientific Reports: raven predation of young tortoises. Detection of *Mycoplasma agassizii* in captive tortoises in Italy, Animals.

Public health and the law

- Centers for Disease Control and Prevention: reptiles and amphibians. Food and Drug Administration: pet turtles, a source of germs. Center for Food Security and Public Health, Iowa State University: reptile-associated salmonellosis.
- 21 CFR 1240.62; Colorado Parks and Wildlife species lists; New Jersey Administrative Code 7:25-4.4 and 4.5; West Virginia Code section 20-2-13; New Mexico Director's Species Importation List; 3 Delaware Administrative Code 903; 321 CMR 9.01, Massachusetts; Hawaii List of Conditionally Approved Animals; New York City Health Code section 161.01; Vermont Unrestricted Wild Animal List; Minnesota Statutes section 97C.605; Montana Code section 87-5-706; Rhode Island Code of Regulations 250-40-05-3; and each other state's rules, read between August and October 2026. Michigan State University College of Law, Animal Legal & Historical Center: pet trusts.

REFERENCE

Sources, continued

Husbandry, equipment and prices.

Husbandry

- ReptiFiles: the Russian tortoise care sheet, and its care guidance as republished by Zen Habitats. Sizes, temperatures, humidity, UVB distances, substrate depth, feeding by age and cleaning.
- Zoo Med Laboratories: the ReptiSun 5.0 and 10.0 UVB T5 HO product pages (tube lengths and wattages).
- Tortoise Trust: captive care of the Russian tortoise, feeding Mediterranean tortoises, and hydration. The wild year, burrows, behavior, sexing, hatchling size and the trade. San Diego Turtle and Tortoise Society: Russian tortoise care sheet. American Society for the Prevention of Cruelty to Animals, Animal Poison Control: buttercup.
- Beastly Facts guides, current at publication: the Russian tortoise enclosure setup, cost, feeding, handling, health issues, enrichment and legal guides, and the cross-species tortoise brumation, soaking, sexing and egg binding, chelonian herpesvirus, safe weeds, outdoor housing, heating and thermostats, UVB lighting, emergency plan, stool and urates, *Salmonella* and hygiene, and succession planning guides. Prices were last checked in September 2026.

REFERENCE

Where the sources disagree

Where credible sources give different numbers, this package prints one and says why.

TOPIC	WHAT THE SOURCES SAY	WHAT THIS PACKAGE USES
Enclosure size	4×2 ft as the indoor minimum (general care guidance); at least 12 sq ft (a care sheet); 7×3.5 ft as the adult minimum, plus 10 sq ft per added tortoise (a specialist care sheet)	4×2 ft as the floor, 12 sq ft at least, 7×3.5 ft as the target
Basking surface	90 to 95°F (a care sheet); 95°F (a specialist care sheet)	About 95°F (35°C), on the surface
Humidity	40 to 50% ambient (a care sheet); 40 to 75% for juveniles, lower by day and higher at night (a specialist care sheet)	40 to 50% at the surface for an adult, 40 to 75% for a juvenile, a moist hide for both
Substrate depth	At least 4 in for a juvenile and 6 in for an adult (a specialist care sheet); 6 to 12 in or more in one dig zone (species enrichment guidance)	4 in juvenile, 6 in adult, a dig zone of 6 to 12 in or more
Substrate change	Every 3 to 6 months (a species care sheet); monthly for a standard enclosure (general hygiene guidance)	Every 3 to 6 months for a deep soil bed, soiled patches as they happen
Adult soak	About once a week for a dry-country adult (veterinary and species care guidance); two to three times a week for at least 20 minutes for indoor tortoises (a specialist charity)	About weekly, 10 to 20 minutes, for an adult; twice a week for a juvenile
UVB tube life	6 to 12 months depending on the model	The maker's schedule, never past 12 months
Brumation length	No more than 10 weeks for a small tortoise and 14 for a large one (a veterinary college); a shorter 6 weeks to 3 months (other veterinary guidance)	The 10 and 14 week ceilings, with shorter always acceptable
Brumation weight loss	Act past 8 to 10% in total (a veterinary college); 6 to 7% acceptable, past 7% a concern (a second veterinary source)	End it past 7%, the more cautious line
Pyramiding	Low humidity during growth (one line of research); the growth rate itself (another)	Both, plus diet: correct humidity, temperature and a slow-growth diet
Fruit	A rare tiny treat for arid tortoises (general tortoise guidance); none at all (a specialist charity and species guidance)	None
Lifespan	40 years or more, some past 50 (species guidance); 40 to 50, with some past 60, a captive figure since wild lifespan is undocumented	40 years or more, some reaching 50 plus

Version history & about

What changed in each edition, so a printed copy can be placed.

Version history

This is edition 3.1 (Oct 2026). The store page shows the current edition; re-download free if yours is older. Where this list and a page disagree, the page wins.

- **3.1, Oct 2026.** Costs: every item priced and rounded to the nearest \$5; setup \$410 to \$835, yearly \$365 to \$715, the budget now over two pages. Other changes are wording only.
- **3.0, Oct 2026.** Rebuilt on the current outline, 51 pages. Safety: basking surface about 95°F (35°C); adult surface humidity 40 to 50%; substrate 4 in juvenile, 6 in adult, dig zone 6 to 12 in; daytime air 75 to 85°F; UVB tube 14 to 16 in through mesh, 17 to 20 in without, replaced every 6 to 12 months, never past 12; growing tortoises fed daily as much as they eat, adults a shell-sized daily portion; multivitamin lightly, no weekly schedule; hatchlings soaked daily 15 to 30 minutes; first vet exam within a week; substrate fully replaced every 3 to 6 months; car trips below 65°F over 30 minutes get a wrapped hand warmer, checked every 20 to 30 minutes. Legal: own section, all 52 jurisdictions and the eleven with stricter local rules. Costs: first checkup about \$75; start-up \$497 to \$1,250; forty years \$14,600 to \$28,600. Other changes are wording only.
- **2.3, Oct 2026.** Safety: enclosure 7×3.5 ft; adults soaked weekly, juveniles twice a week, never unattended; vitamin D3 only when a vet asks; no fruit; pen walls about 2 ft with an inward lip, buried 12 in. Legal: West Virginia added, so 5 states require a permit.
- **2.2, Sep 2026.** Legal: Colorado bans the species, and more states were found to require a permit or attach a condition.
- **2.1, Sep 2026.** Safety: brumation at 35 to 50°F, ideally 41°F, for at most 10 to 14 weeks by size, after a 1 to 3 week fast at 70 to 80°F, and not under about 4 years old; quarantine six months with testing; medicine doses removed.
- **2.0, Sep 2026.** Rebuilt, 37 pages. Replaced by 3.0.
- **1.0, Aug 2026.** First edition, 21 pages. Replaced by 2.0.

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a reptile-experienced veterinarian, and nothing in it should delay a vet visit. Treatments described here are for recognizing what a vet may do, never for use at home. The law pages are a summary as of October 2026, not legal advice.

© Beastly Facts · Edition 3.1 · October 2026 · For personal use by the purchaser. Please don't redistribute.